

Sangwoo **Kim**

Researcher at KETI

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"I enjoy turning theory into systems that work in the real world."

Research Experience

Korea Electronics Technology Institute (KETI) — *South Korea*

Researcher, Autonomous Manufacturing Research Center — *Sep. 2026 - Present*

- Participate in Subproject 1 of *Development of Core Foundation Model and AI Agent Technology for Thermal-Process Manufacturing (TP-MFM)*, focusing on AI agent research.
- Contribute to first-year thermal-process agent design spanning context interpretation, task generation and scheduling, foundation-model routing, and feedback-based policy refinement for operator-approved operating-condition recommendations.
- Review industrial reinforcement-learning studies to inform control-policy design, state and KPI definitions, physical constraints, and requirements for offline-learning and virtual-testbed evaluation.

Research Interests

- **Multi-Agent Reinforcement Learning (MARL)** — Studying how learning agents can coordinate decisions across coupled manufacturing processes. Motivated by my thermal-process AI-agent research, I want to investigate cooperation, credit assignment, and the balance between local objectives and plant-level constraints under changing operating conditions.
- **Reinforcement Learning for Manufacturing Control** — Studying data-efficient policies that balance temperature accuracy, thermal uniformity, and heat input within operating limits. Building on my industrial-RL literature review and policy-design work, I am interested in offline learning, simulation-based evaluation, and constraint-aware policy improvement for thermal processes.
- **Industrial AI Agents** — Connecting LLM-based context interpretation and task planning with foundation-model routing, prediction, and specialized control modules. Extending my KETI agent-design work and NOVA, I want to study reliable tool use and feedback, with operating-condition recommendations subject to explicit safety checks and operator approval.

Education

Chung-Ang University — *Seoul, S.Korea*

B.S. in Civil & Environmental Eng. and Electrical & Electronic Eng. (Double Major) — *Mar. 2021 - Aug. 2026*

Hongik University — *Sejong, S.Korea*

Undergraduate Studies — *Mar. 2019 - Feb. 2021*

- Transferred to Chung-Ang University.

Selected Projects

NOVA — Evidence-Grounded Financial Research Assistant

Independent project; AI-assisted development — *May 2026 - Jul. 2026*

- Built a financial fact-checking and question-answering service that retrieves evidence from DART disclosures and uploaded PDFs and uses external tools to check claims.
- Implemented user-scoped document retrieval with pgvector and a FastAPI backend, connected to Next.js and SwiftUI clients.
- Improved cached fact-checking by correcting cache lookups and parallelizing company synchronization and claim verification.

Pi-FiRi — Fire-Monitoring Robot with an LLM Agent

ToBig's 20th Conference; four-person team — *May 2025 - Jul. 2025*

- Designed the distributed Raspberry Pi / PC inference architecture and LLM reporting integration for a fire-monitoring robot; collaborated on model training and robot control.
- Integrated EfficientDet-Lite1 fire/smoke detection and continuous-detection filtering with structured patrol logs and four-tier alerts.

Selected Projects (continued)

Curat3R — Single-Image 3D Reconstruction Archiving Platform

ToBig's 21st Conference; five-person team — *Oct. 2025 - Jan. 2026*

- Contributed the input-suitability filter and system architecture for a platform that turns a single image into an interactive 3D asset.
- Implemented a CLIP ViT-B/32 filter with six categories and an accept/reject decision, with a human override for input selection.
- Connected SPAR3D previews and TRELLIS.2 reconstruction to a Three.js gallery for interactive inspection and archiving.

Additional Projects

Smart-Warehouse Shipping-Delay Prediction

DACON; independent project — *Apr. 2026 - May 2026*

- Reduced validation MAE from 12.616 to 9.989 using a four-model ensemble of LightGBM, CatBoost, BiGRU, and Attention GRU; built a Dash dashboard for scenario-based batch predictions.

Menu Demand Forecasting

LG Aimers 7th; four-person team — *Aug. 2025 - Sep. 2025*

- Focused on loss-function design and rolling validation for CNN–BiLSTM menu-demand forecasts; team advanced to the offline finals and recorded 5th place.

Steel Quality Prediction

MixUP AI Datathon; three-person team — *May 2025*

- Contributed EDA, preprocessing, and LightGBM modeling for OK/NG classification; used RFECV plus a process-derived feature for a final seven-feature model and SHAP for interpretation.

Prompt-Based Bias Mitigation

DACON Bias-A-Thon, Track 2; team project — *Apr. 2025 - May 2025*

- Designed context-grounded two-stage prompts without fine-tuning and regex-based answer extraction for consistent response formatting.

Activities

ToBig's — AI & Big Data Analysis Club

Operations Manager & Member, 23rd Cohort — *Jan. 2025 - Jan. 2026*

- Organized study groups and team projects; co-designed a vibration/acoustic leak-level regression track and Kaggle-based evaluation for the MixUP AI datathon.

MACH Aerospace Club, Chung-Ang University

Member, Drone Control Team — *Mar. 2024 - Sep. 2024*

- Worked on embedded drone gimbal and landing-gear control using Raspberry Pi, OpenCV, and HC-12 RF communication.

Competition Results & Honors

2026 **Top 11% (68th)**, DACON — Smart-Warehouse Shipping-Delay Prediction
2025 **5th Place (Finals)**, LG Aimers 7th — Offline Finals, Menu Demand Forecasting
2025 **5th Place**, MixUP AI Datathon — Steel Quality Prediction
2024 **Excellence Award**, ALCP Self-Directed Learning Award, Chung-Ang University

Certifications

Jul. 2026 **Big Data Analysis Engineer**, Korea Data Agency (K-DATA)

Skills

ML / Agents Time-Series Modeling, LLM Agents, Tool Calling, RAG, Computer Vision, 3D Reconstruction
Frameworks PyTorch, TensorFlow, scikit-learn, LightGBM, OpenCV, Pandas
Programming Python, C++
Systems / Edge FastAPI, PostgreSQL / pgvector, Git, Raspberry Pi, Arduino
Languages Korean (Native), English